Contributions — who did what

Team: Paridhi Mittal and her teammate (two-person team, as the brief allows) **Sources for every row:** the git history (`git log --format="%h %an %ad %s"`), the teammate's own notes in her commit (`docs/FOR_PARIDHI.md` at `957224c`), the first implementation kept in `legacy/`, and the diff between the two implementations. Each person's AI assistance is disclosed in `AI_USE.md`.

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1. At a glance

Area	Paridhi	Teammate
Frozen data snapshot, checksum, provenance	**built it** (11 Oct)	built on it
Dataset audit	**missingness, monthly counts, numeric anomalies, field inspection** (11 Oct)	verified and characterised five defects with measurements
Design note	**wrote it**: six agents with Reads/Writes, both cycles, `query_mode`, MR table, LangGraph	reviewed it against measurements; proposed corrections
Agent contracts	**13-file suite** (v1); rewrote it as the final 12-file suite (v2.1)	—
Data, audit and cleaning scripts	**31 scripts** (v1)	—
First implementation	**criteria, retrieval, anomaly agents, execution graph, typed schemas, tests** (v1)	—
Second implementation (LangGraph graph, statistics, dashboard)	—	**built it**, following the design note
Metamorphic relations and harness	added M4 and 36 radius inputs; base-input statistics; rejected-count recording	**11 relations, harness, base inputs**
Synthetic control snapshot	repaired the generator	**built it**, with its oracle
Local-model mode	safeguards, I3 review fix, narrative fixes, offline replay	**built it**: Ollama client, prompt-hygiene rules, determinism probe
Pre-v2 sweeps and diagnosis	—	**ran them** (deterministic and two LLM sweeps on Kaggle); Findings 1–9
Merge of the two implementations	**did it**, in reviewed phases	—
Data rules and validation in the final system	**ported from v1**: IST months, Price-on-Request, price validity, unique IDs, request validation	—
Alignment with the refined design note (v2.1)	**did it**; documented ten deviations	—
v2 sweeps (deterministic, synthetic, LLM)	**ran them**; LLM sweep on Kaggle; verified by offline replay	—
Faults found by the relations in v2	**M3** (widening), **I3** (LLM review), narrative guard — each fixed and re-measured	—
RESIDEX cross-check, threshold sensitivity	made the sensitivity table reproducible (script)	**did the cross-check**; first table
Report	rewrote for v2; cut to 10 pages; PDF	wrote the first report (pre-v2)
NOTEBOOK, AI_USE, DIAGNOSIS	Stages 16–24; AI_USE §§5–6; Findings 10–11; issue index	Stages 0–15; AI_USE §§1–4; Findings 1–9
Tests	84 functions (v1); final suite 132 functions	84 functions (v2 starting point)
Commits	all the others: 3 for the foundation, 1 for the first implementation (`459aeca`), the rest for the integration	2 (each a large snapshot of her work)

Commit counts are not a measure of effort: the teammate committed her work in two large snapshots, and Paridhi's first implementation was committed in one.

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2. Timeline

Date	Who	What	Commits
11 Oct	Paridhi	froze the raw snapshot with checksum and Readme	`0caeb2e`
11 Oct	Paridhi	dataset audit; field and anomaly inspection	`3793127`, `273045e`
≤ 12 Oct	Paridhi	design note; first implementation, contracts, scripts, tests (six logged test runs on 12 Oct)	in `459aeca`; `legacy/`
14–15 Oct	Teammate	second implementation, built from `273045e` and aligned to the design note; pre-v2 sweeps; diagnosis; first report	`736d594`, `957224c`
15 Oct	Paridhi	merge; contract suite v2; data rules and validation ported; harness and agent fixes; M3 fault found and fixed; v2 sweeps	`bed240b` … `65db1ec`
15 Oct	Paridhi	alignment with the refined design note (v2.1); re-run	`f4752f8` … `5b9271e`
16 Oct	Paridhi	v2 LLM sweep on Kaggle; I3 fault found and fixed; offline replay; report cut; logs and issue index	`c8e6e85` … `a2988b5`

3. Paridhi's work

3.1 Foundation (before the teammate's work)

- **Frozen snapshot** (`0caeb2e`): the 99acres Hyderabad slice, 9,487 rows × 55 columns, with its SHA-256 checksum and a provenance Readme. The dataset citation and licence were later verified as correct, against a suggestion that they were a transcription error (`docs/FOR_PARIDHI.md`, item 10).
- **Dataset audit** (`3793127`, `273045e`): missingness per column, monthly listing counts, numeric anomalies, field inspection (`audit/hyderabad_*.csv`, `scripts/audit_hyderabad.py`, `scripts/inspect_hyderabad_*.py`).
- **Design note** (`Assignment1_Design_Note_Final.docx`): the six agents with explicit Reads/Writes; cycle 2 as Critique → Analysis with a specific complaint and flagged months; `query_mode` as the genuine dependency; the control-flow table with "why necessary"; the five-column MR table; LangGraph. The teammate's notes record adopting all of these, and that cycle 2 and `query_mode` replaced her earlier design.
- **Contract suite** (`legacy/contracts/`, 13 files, about 33,000 words): global contract, data, agents, planner, validator, anomaly, reporter, aggregator, diagnostic, LLM, testing, golden, and a consistency audit.
- **Data, audit and cleaning scripts** (`legacy/scripts/`, 31 scripts): the cleaning pipeline and processed snapshot, the Price-on-Request exclusion, reporting-month identity, locality / price / month audits, runtime-state builders.
- **First implementation** (`legacy/app/`, about 3,250 lines): criteria agent, retrieval agent, anomaly detector, execution graph, typed schemas, snapshot repository.
- **Tests** (`legacy/tests/`): 84 test functions in 15 files, with six logged runs (`legacy/audit/test_logs/`; last: 77 passed, 12 subtests).

3.2 Integration and the final system (after the teammate's work)

Detailed change by change, with reasons, in `docs/INTEGRATION_REPORT.md`. In summary:

- **Merged** the second implementation in reviewed phases; made the snapshot checksum survive Windows; moved the first implementation to `legacy/`.
- **Contracts:** rewrote the suite for the integrated system (12 files, v2.1.0-draft), with ten argued deviations from the refined design note.
- **Data rules** ported from the first implementation: months in IST, Price-on-Request and price-validity flags (rows kept, not dropped), unique listing IDs, locality IDs, the midpoint listing price. Population 4,975 listings, 4,574 with a usable price.
- **Request validation** ported (`src/validation.py`) and wired into the dashboard.

- **Agents:** coverage with ACCEPT / RETRY / LIMITED and the note's lever order; geography widening that keeps the locality filter and provably preserves set inclusion; critique coherence checks and NEEDS_REVIEW; correct rejection accounting; provenance of every report.
- **Harness:** 36 radius base inputs and relation M4 (72 inputs, 12 relations); the base input as the statistical unit; rejected counts recorded per run; transforms that keep price fields consistent.
- **Local model:** weights digest recorded; a check that the model answers before a sweep; stop on a dead server; replies saved as they arrive; the offline replay that reproduces the LLM sweep without a GPU.
- **Sweeps:** the v2 deterministic and synthetic sweeps, and the v2 LLM sweep on a Kaggle T4 — run, re-run after the fix, and verified by replay.
- **Faults found by the relations and fixed:**
- **M3**: the second implementation's widening dropped the locality filter, so M3 could not fail; Paridhi's first fix then broke set inclusion, M3 caught it (9/36), and the final rule fixed it (0/36);
- **I3 under the model**: the review cut its batch before removing duplicates — a line from the second implementation, also behind its open I3 residual (8/72 → 0/72);
- the **narrative guard** rejected figures it had itself supplied, and the model copied the facts when there was no price change.
- **Reproducibility:** headline-table checks for both sweeps, `verify_claims.py` for the v2 population, the threshold-sensitivity script, golden regeneration with recorded reasons, logged check runs, a clean-checkout check.
- **Documents:** the report rewritten for v2 and cut to 10 pages (with PDF); NOTEBOOK Stages 16–24; AI_USE §§5–6; DIAGNOSIS Findings 10–11; the issue index; the integration report.
- **Tests:** from 84 to 132 test functions (154 cases), including request validation and a dashboard test.

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4. The teammate's work

All in commits `736d594` and `957224c` (14–15 October), built from Paridhi's commit `273045e` and aligned to her design note:

- **The second implementation:** the LangGraph graph with two cycles and four parallel retrieval agents with stable hashing (`src/agents/`, 1,254 lines of agent code among 3,592 lines in `src/`); the loader, output contract and statistics; the dashboard (`app.py`, `src/viz.py`).
- **Defect verification:** measured the five data defects — the square-metre area column (ratio 10.7639 at both tails), the int32 sentinel, mixed sale / rent / PG listings, structurally missing bedrooms on land, area-entry errors — and kept the last for cycle 2.
- **Metamorphic relations and harness:** 11 relations across the three families, 36 base inputs, three seeds, Wilson intervals, control-path stability, the retained counter-example `M1raw`.
- **Synthetic control snapshot** with a known trend and its oracle.
- **Local-model mode:** the Ollama client with seeding, caching and counted fallbacks; the four prompt-hygiene rules; the determinism probe (20/20 identical replies); two LLM sweeps on Kaggle.
- **Diagnosis:** Findings 1–9 — the per-worker outlier threshold behind I3 and I4, M1's band crossing, the non-idempotent filter behind M2, the synthetic-only loop tension, a harness bug, the prompt faults and the architectural trade-off.
- **External cross-check:** NHB RESIDEX against the snapshot, and the threshold-sensitivity table showing the headline's sign depends on the threshold.
- **Scripts:** `run_harness.py`, `diagnose.py`, `headline_table.py`, `verify_claims.py`, `prompt_sensitivity.py`, `make_synthetic.py`, `make_results_section.py`, `validate_llm.py`.
- **Tests:** 84 test functions (101 cases) in 7 files.
- **Documents:** the first REPORT, DESIGN (structured to mirror the design note), NOTEBOOK Stages 0–15, AI_USE §§1–4, DIAGNOSIS.

- **Review of the design note** (`docs/FOR_PARIDHI.md`): 25 items, including five measured corrections — the sufficiency flag, renaming days on market, M1's band-free subdomain, D2's tolerance, and the conditionality of M2 / M3.

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5. Where each part of the final system came from

Component	Origin	Final state
Snapshot and checksum	Paridhi	unchanged; pinned on every load
Architecture: six agents, Reads/Writes, both cycles, `query_mode`	Paridhi's design note	implemented by the teammate; aligned to the refined note by Paridhi
LangGraph graph, parallel retrieval	teammate	kept
Data rules: IST months, Price on Request	Paridhi (v1)	ported by Paridhi
Data rules: price-validity flags, unique listing IDs, midpoint price	Paridhi (integration)	new
Defect handling (square metres, sentinels, sale only, dwellings)	teammate's measurements	kept
Output contract	teammate (v1 fields); refined note	rewritten by Paridhi (v2.1)
Contracts	Paridhi (v1 format)	rewritten by Paridhi
Request validation	Paridhi (v1)	ported by Paridhi
Relations I1–I5, M1–M3, M1raw, D1, D2	teammate	updated by Paridhi for the new contract
M4 and radius inputs	Paridhi	new
Harness statistics (base-input unit)	teammate (per run)	changed by Paridhi
Synthetic control	teammate	repaired by Paridhi
LLM mode and prompt hygiene	teammate	safeguards and two fixes by Paridhi
Pre-v2 diagnosis	teammate	kept in `docs/DIAGNOSIS.md`
v2 sweeps and diagnosis	Paridhi	the report's results
Report	teammate (pre-v2)	rewritten by Paridhi

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6. AI assistance

Both implementations were built with AI assistance, as the brief encourages: the teammate's with Claude through the Claude Code CLI (`AI_USE.md` §1), and Paridhi's integration with Claude Opus 5.5 through the Claude Code desktop app (`AI_USE.md` §§1, 5–6), including one commit (`2edd7bf`) from a second session she ran in parallel. In both cases the decisions, the runs and the responsibility for the results were the authors'.